CHANG CHIRUI

EDUCATION

The University of Hong Kong, CVMI LAB — Hong Kong
Ph.D. in Computer Vision — *Sept. 2023 – Present*
Supervised by Prof. Xiaojuan Qi

The Chinese University of Hong Kong — Hong Kong
B.Sc. in Computer Science — *Aug. 2018 – Jul. 2022*

RESEARCH INTERESTS

Research at the intersection of video generation, 3D/world modeling, and graphics- and simulation-driven data generation, with a focus on controllable video world synthesis and 3D-aware generation/editing.

RESEARCH/WORK EXPERIENCE

Tencent — Shenzhen, China
Project UP Research Internship
State-aware streaming video generation — *June 2026 – Present*

Kling, Kuaishou Technology — Shenzhen, China
Research Internship, Base Model Team
3D consistency in video generation models, Unified video editing — *Jul. 2025 – May 2026*

PUBLICATIONS/MANUSCRIPTS

- State Forcing: Telic Instruction Following for Streaming Video Generation
  Chirui Chang, Fuchen Long, Xinlong Sun, Xi Chen, Xiaojuan Qi
  Under Review, 2026
- Geometry-Instructed Video Editing
  Chirui Chang, Xiaoyang Lyu, Yi-Hua Huang, …, Yikang Ding, Jianmin Bao, Xin Tao, Pengfei Wan, Xiaojuan Qi
  ACM SIGGRAPH Asia, 2026
- How Far are AI-generated Videos from Simulating the 3D Visual World: A Learned 3D Evaluation Approach
  Chirui Chang, Jiahui Liu, Zhengzhe Liu, Xiaoyang Lyu, Yi-Hua Huang, Xin Tao, Pengfei Wan, Di Zhang, Xiaojuan Qi
  IEEE/CVF International Conference on Computer Vision (ICCV), 2025
- Total-Decom: Decomposed 3D Scene Reconstruction with Minimal Interaction
  Xiaoyang Lyu*, **Chirui Chang***, Peng Dai, Yang-Tian Sun, Xiaojuan Qi
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
- MarS3D: A Plug-and-Play Motion-Aware Model for Semantic Segmentation on Multi-Scan 3D Point Clouds
  Jiahui Liu*, **Chirui Chang***, Jianhui Liu, Xiaoyang Wu, Lan Ma, Xiaojuan Qi
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- SynUrbanSAT: Controllable Satellite Data Synthesis for Urban Semantic 3D Reconstruction
  Jun Jiang, Yixuan Ma, …, **Chirui Chang**, …, Donglin Yang, Xuguang Wang
  Under Review, 2026

- From Prediction to Control: A Survey of Closed-Loop World Models for Robot Learning
  Hejia Geng, Junzhe Wang, …, **Chirui Chang**, …, Yingcheng Wu, Zhengfei Yin
  Under Review, 2026
- AniGen: Unified S^3 Fields for Animatable 3D Asset Generation
  Yi-Hua Huang, Zi-Xin Zou, Yuting He, **Chirui Chang**, …, Yuan-Chen Guo, Yan-Pei Cao, Xiaojuan Qi
  ACM SIGGRAPH, 2026
- ObjectMorpher: 3D-Aware Image Editing via Deformable 3DGS
  Yuhuan Xie, Aoxuan Pan, Yi-Hua Huang, **Chirui Chang**, Peng Dai, Xin Yu, Xiaojuan Qi
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- ASSIST-3D: Adapted Scene Synthesis for Class-Agnostic 3D Instance Segmentation
  Shengchao Zhou, Jiehong Lin, Jiahui Liu, Shizhen Zhao, **Chirui Chang**, Xiaojuan Qi
  AAAI Conference on Artificial Intelligence (AAAI), 2026
- Dynamic Important Example Mining for Reinforcement Finetuning
  Haoru Tan, Sitong Wu, …, **Chirui Chang**, Shaofeng Zhang, Samm Sun, Xiuzhe Wu, Ruobing Xie, Xiaojuan Qi
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Dataset Distillation by Influence Matching
  Haoru Tan, Wang Wang, Sitong Wu, Xiuzhe Wu, Yang-Tian Sun, **Chirui Chang**, Shaofeng Zhang, Xiaojuan Qi
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Aligning Effective Tokens with Video Anomaly in Large Language Models
  Yingxian Chen, Jiahui Liu, Ruidi Fan, Yanwei Li, **Chirui Chang**, Shizhen Zhao, Wilton W.T. Fok, Xiaojuan Qi, Yik-Chung Wu
  IEEE/CVF International Conference on Computer Vision (ICCV), 2025
- NoteIt: A System Converting Instructional Videos to Interactable Notes Through Multimodal Video Understanding
  Running Zhao, Zhihan Jiang, Xinchen Zhang, **Chirui Chang**, Handi Chen, Weipeng Deng, Luyao Jin, Xiaojuan Qi, Xun Qian, Edith C.H. Ngai
  ACM Symposium on User Interface Software and Technology (UIST), 2025
- Object-level scene deocclusion
  Zhengzhe Liu, Qing Liu, **Chirui Chang**, Jianming Zhang, Daniil Pakhomov, Haitian Zheng, Zhe Lin, Daniel Cohen-Or, Chi-Wing Fu
  ACM SIGGRAPH, 2024

∗ denotes co-first author with equal contribution(coin-toss order).

HONORS

- Postgraduate Scholarship (PGS), HKU
- S.H.Ho College Master's List Scholarship, S.H.Ho college, CUHK
- CSE Outstanding Academic Performance, CSE Department, CUHK
- Elite Stream Scholarship, Engineering Faculty, CUHK
- Admission Scholarship, S.H.Ho college, CUHK
- Winners of Professor Charles K. Kao Student Creativity Awards 2021